The multivariate Gaussian distribution

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The multivariate Gaussian distribution

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Covariance matrices

Let $\boldsymbol{X} = (X_1, X_2, \ldots, X_n)^t$ be an n-dimensional random vector with expectation $\mathbb{E}(\boldsymbol{X}) = \boldsymbol{m}_X = (m_{X_1}, m_{X_2}, \ldots, m_{X_n})^t$.

Definition

The covariance matrix of $\boldsymbol{X}$ is the $n \times n$ matrix $\boldsymbol{K}_X = (k_{ij})$ defined as follows:

$$\boldsymbol{K}_X = \mathbb{E}\left((\boldsymbol{X} - \boldsymbol{m}_X)(\boldsymbol{X} - \boldsymbol{m}_X)^t\right)$$

Therefore,

- *If $i \neq j$, then $k_{ij} = \mathbb{E}\left((X_i - m_{X_i})(X_j - m_{X_j})\right) = \mathrm{Cov}(X_i, X_j)$.*
- *The diagonal entries are $k_{ii} = \mathbb{E}\left((X_i - m_{X_i})^2\right) = \mathrm{Var}(X_i)$.*

Covariance matrices

Proposition

The covariance matrix $\boldsymbol{K}_X$ is:

- *symmetric,* $k_{ij} = \mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i) = k_{ji}$.
- *positive-semidefinite; that is to say,* $\boldsymbol{z}^t \boldsymbol{K}_X \boldsymbol{z} \geqslant 0$ *for all* $\boldsymbol{z} = (z_1, z_2, \ldots, z_n)^t \in \mathbb{R}^n$.

Proof: Let $Y = z_1 X_1 + z_2 X_2 + \cdots + z_n X_n = \boldsymbol{z}^t \boldsymbol{X}$. Observe that

$$m_Y = \mathbb{E}\left(\sum_{i=1}^{n} z_i X_i\right) = \sum_{i=1}^{n} z_i m_{X_i} = \boldsymbol{z}^t \boldsymbol{m}_X$$

Therefore, $Y - m_Y = \boldsymbol{z}^t(\boldsymbol{X} - \boldsymbol{m}_X)$.

Covariance matrices

Thus,

$$\begin{aligned}
\boldsymbol{z}^t \boldsymbol{K}_X \boldsymbol{z} &= \boldsymbol{z}^t \, \mathbb{E}\left((\boldsymbol{X} - \boldsymbol{m}_X)(\boldsymbol{X} - \boldsymbol{m}_X)^t \right) \boldsymbol{z} \\
&= \mathbb{E}\left(\boldsymbol{z}^t \, (\boldsymbol{X} - \boldsymbol{m}_X)(\boldsymbol{X} - \boldsymbol{m}_X)^t \boldsymbol{z} \right) \\
&= \mathbb{E}\left((Y - m_Y)(Y - m_Y)^t \right) \\
&= \mathbb{E}\left((Y - m_Y)^2 \right) = \mathrm{Var}(Y) \geqslant 0
\end{aligned}$$

Covariance matrices

Let us say that the "centred" random variables

$$X_1 - m_{X_1},\ X_2 - m_{X_2},\ \ldots,\ X_n - m_{X_n}$$

are linearly independent (in the sense of linear algebra) if the equality

$$\sum_{i=0}^{n} z_i \left(X_i - m_{X_i}\right) = 0 \quad \text{(with probability 1)}$$

implies that $z_1 = z_2 = \cdots = z_n = 0$.

Covariance matrices

Theorem

The random variables

$$X_1 - m_{X_1},\ X_2 - m_{X_2},\ \ldots,\ X_n - m_{X_n}$$

are linearly independent if and only if the covariance matrix $\boldsymbol{K}_X$ *is positive-definite; that is, if and only if*

$$\boldsymbol{z}^t \boldsymbol{K}_X \boldsymbol{z} > 0 \quad \textit{for all } \boldsymbol{z} \neq \boldsymbol{0}.$$

Covariance matrices

Proof: Let $Y = \boldsymbol{z}^t \boldsymbol{X}$ and observe that

$$\boldsymbol{z}^t \boldsymbol{K}_X \boldsymbol{z} = 0 \text{ for some } \boldsymbol{z} \neq \boldsymbol{0}$$

$$\iff \mathrm{Var}(Y) = 0 \text{ for some } \boldsymbol{z} \neq \boldsymbol{0}$$

$$\iff Y - m_Y = 0 \quad \text{(with probability 1) for some } \boldsymbol{z} \neq \boldsymbol{0}$$

$$\iff \sum_{i=1}^{n} z_i \left(X_i - m_{X_i}\right) = 0 \quad \text{(with probability 1)}$$

$$\text{for some } \boldsymbol{z} = (z_1, z_2, \ldots, z_n)^t \neq \boldsymbol{0}$$

$$\iff X_1 - m_{X_1}, X_2 - m_{X_2}, \ldots, X_n - m_{X_n}$$

$$\text{are not linealy independent.}$$

Linear transformations

Theorem

Let $\boldsymbol{X} = (X_1, X_2, \ldots, X_n)^t$ *be an n-dimensional random vector,* $\boldsymbol{A}$ *an* $m \times n$ *real matrix, and* $\boldsymbol{b}$ *a constant real* $m \times 1$ *vector. Define the m-dimensional random vector* $\boldsymbol{Y} = (Y_1, Y_2, \ldots, Y_m)^t$ *as follows:*

$$\boldsymbol{Y} = \boldsymbol{A}\,\boldsymbol{X} + \boldsymbol{b}$$

The expectation vector and covariance matrix of $\boldsymbol{Y}$ *are given by:*

$$\boldsymbol{m}_Y = \boldsymbol{A}\,\boldsymbol{m}_X + \boldsymbol{b}, \quad \boldsymbol{K}_Y = \boldsymbol{A}\,\boldsymbol{K}_X\,\boldsymbol{A}^t$$

Observe that if $\boldsymbol{A} = (a_{ij})$ and $\boldsymbol{b} = (b_i)$, then

$$Y_i = a_{i1}X_1 + a_{i2}X_2 + \cdots + a_{in}X_n + b_i, \quad 1 \leqslant i \leqslant m$$

Linear transformations

Proof: By the linearity of the expectation operator we have

$$\boldsymbol{m}_Y = \mathbb{E}(\boldsymbol{Y}) = \mathbb{E}(\boldsymbol{A}\boldsymbol{X} + \boldsymbol{b}) = \boldsymbol{A}\,\mathbb{E}(\boldsymbol{X}) + \boldsymbol{b} = \boldsymbol{A}\boldsymbol{m}_X + \boldsymbol{b}$$

Analogously,

$$\begin{aligned}
\boldsymbol{K}_Y &= \mathbb{E}\left((\boldsymbol{Y} - \boldsymbol{m}_Y)(\boldsymbol{Y} - \boldsymbol{m}_Y)^t\right) \\
&= \mathbb{E}\left(\boldsymbol{A}\,(\boldsymbol{X} - \boldsymbol{m}_X)(\boldsymbol{X} - \boldsymbol{m}_X)^t\,\boldsymbol{A}^t\right) \\
&= \boldsymbol{A}\,\mathbb{E}\left((\boldsymbol{X} - \boldsymbol{m}_X)(\boldsymbol{X} - \boldsymbol{m}_X)^t\right)\boldsymbol{A}^t \\
&= \boldsymbol{A}\boldsymbol{K}_X\boldsymbol{A}^t
\end{aligned}$$

Gaussian characteristic functions

Let X_1, X_2, ..., X_n be independent Gaussian random variables, $X_i \sim \mathrm{N}(m_{X_i}, \sigma^2_{X_i})$. Their joint characteristic function is

$$\begin{aligned} M_X(\omega_1, \omega_2, \ldots, \omega_n) &= M_{X_1}(\omega_1) M_{X_2}(\omega_2) \cdots M_{X_n}(\omega_n) \\ &= \prod_{i=1}^{n} \exp\left(i\omega_i m_{X_i} - \frac{1}{2}\sigma^2_{X_i}\omega_i^2 \right) \\ &= \exp\left(\sum_{i=1}^{n} \left(i\omega_i m_{X_i} - \frac{1}{2}\sigma^2_{X_i}\omega_i^2 \right) \right) \\ &= \exp\left(i\boldsymbol{\omega}^t \boldsymbol{m}_X - \frac{1}{2}\boldsymbol{\omega}^t \boldsymbol{K}_X \boldsymbol{\omega} \right) \end{aligned}$$

Gaussian characteristic functions

- In the previous expression, $\boldsymbol{\omega} = (\omega_1, \omega_2, \ldots, \omega_n)^t$ and $\boldsymbol{m}_X = (m_{X_1}, m_{X_2}, \ldots, m_{X_n})^t$ is the expectation vector.
- Moreover,

$$\boldsymbol{K}_X = \begin{pmatrix} \sigma^2_{X_1} & 0 & . & . & . & 0 \\ 0 & \sigma^2_{X_2} & 0 & . & . & 0 \\ & & . & & & \\ & & & . & & \\ 0 & . & . & . & 0 & \sigma^2_{X_n} \end{pmatrix}$$

is the covariance matrix. $\boldsymbol{K}_X$ is diagonal because the random variables X_i are independent. This means that $\mathrm{Cov}(X_i, X_j) = 0$ whenever $i \neq j$.

Gaussian random vectors

Definition

An n-dimensional random vector $\boldsymbol{X} = (X_1, X_2, \ldots, X_n)^t$ is considered Gaussian if its characteristic function takes the form

$$M_X(\omega_1, \omega_2, \ldots, \omega_n) = \exp\left(i\boldsymbol{\omega}^t\boldsymbol{m} - \frac{1}{2}\boldsymbol{\omega}^t\boldsymbol{K}\boldsymbol{\omega}\right),$$

where $\boldsymbol{\omega}^t = (\omega_1, \omega_2, \ldots, \omega_n)$, $\boldsymbol{m}$ is a column $n \times 1$ vector of real numbers, and $\boldsymbol{K}$ is a symmetric positive-semidefinite $n \times n$ matrix.

We denote $\boldsymbol{X} \sim \mathrm{N}(\boldsymbol{m}, \boldsymbol{K})$ to indicate that the random variables X_1, X_2, ..., X_n are jointly Gaussian.

Marginal distributions

If $\boldsymbol{m} = (m_j)$ and $\boldsymbol{K} = (k_{ij})$, then

$$M_{X_j}(\omega) = M_X(0, \ldots, \omega, \ldots, 0) = \exp\left(i m_j \, \omega - \frac{1}{2} k_{jj} \, \omega^2\right)$$

- Since $\boldsymbol{K}$ is positive-semidefinite, $k_{jj} \geqslant 0$ for all j.

If $k_{jj} > 0$, the component X_j is a 1-dimensional Gaussian random variable with parameters

$$\mathbb{E}(X_j) = m_j, \quad \mathrm{Var}(X_j) = k_{jj}$$

Marginal distributions

- If $k_{jj} = 0$ for some j, then

$$M_{X_j}(\omega) = e^{im_j\omega}$$

Hence, $\mathbb{P}(X_j = m_j) = 1$ and so

$$X_j = m_j, \quad \text{(with probability 1)}$$

We can regard the constant variable X_j as a degenerate $\mathrm{N}(m_j, 0)$ Gaussian random variable.

Marginal distributions

Moreover,

$$m_{11;X_rX_s} = \frac{1}{i^2} \left. \frac{\partial^2 M_X(\omega_1, \omega_2, \ldots, \omega_n)}{\partial\omega_r \, \partial\omega_s} \right|_{(0,0,\ldots,0)} = k_{rs} + m_r m_s$$

Therefore,

$$\mathrm{Cov}(X_r, X_s) = k_{rs}$$

- So we conclude that $\boldsymbol{K}$ is the covariance matrix of $\boldsymbol{X}$.

Eigenvalues of the covariance matrix

Given that the matrix $\boldsymbol{K}_X$ is symmetric, it can be diagonalised through an orthogonal transformation.

Consequently, there exists an orthogonal matrix $\boldsymbol{C}$ such that $\boldsymbol{C}\boldsymbol{C}^t = \boldsymbol{C}^t\boldsymbol{C} = \boldsymbol{I}$, and for which

$$\boldsymbol{C}\boldsymbol{K}_X\boldsymbol{C}^t = \boldsymbol{D} = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n),$$

where the numbers $\lambda_i \in \mathbb{R}$ are the eigenvalues of $\boldsymbol{K}_X$.

Equivalently,

$$\boldsymbol{K}_X = \boldsymbol{C}^t\boldsymbol{D}\boldsymbol{C}$$

Eigenvalues of the covariance matrix

Hence,

$$0 \leqslant \boldsymbol{z}^t \boldsymbol{K}_X \boldsymbol{z} = \boldsymbol{z}^t (\boldsymbol{C}^t \boldsymbol{D} \boldsymbol{C}) \boldsymbol{z}$$
$$= (\boldsymbol{C}\boldsymbol{z})^t \boldsymbol{D} (\boldsymbol{C}\boldsymbol{z}) = \boldsymbol{y}^t \boldsymbol{D} \boldsymbol{y} = \sum_{i=1}^{n} \lambda_i y_i^2,$$

where $\boldsymbol{y} = \boldsymbol{C}\boldsymbol{z} \in \mathbb{R}^n$ can be arbitrarily chosen, because $\boldsymbol{z} = \boldsymbol{C}^t \boldsymbol{y}$.

We deduce that:

- All the eigenvalues of $\boldsymbol{K}_X$ are nonnegative, $\lambda_i \geqslant 0$ for all $1 \leqslant i \leqslant n$.
- The covariance matrix $\boldsymbol{K}_X$ is positive-definite if and only if $\lambda_i > 0$ for all $1 \leqslant i \leqslant n$.

Linear independence of the components

Proposition

Let $\boldsymbol{X}$ be a Gaussian random vector. Then the covariance matrix $\boldsymbol{K}_X$ is positive-definite if and only if any of the following statements hold:

- $\lambda_i > 0$ *for all* $1 \leqslant i \leqslant n$.
- *The variables* $X_i - m_{X_i}$, $1 \leqslant i \leqslant n$, *are linearly independent.*
- $\det(\boldsymbol{K}_X) > 0$.
- $\boldsymbol{K}_X$ *has an inverse.*
- *The joint probability density function,* $f_X(x_1, x_2, \ldots, x_n)$, *exists.*

Linear independence of the components

(Remark)

Gaussian random vectors remain well-defined even if the covariance matrix $\boldsymbol{K}_X$ *is not invertible, that is, when* $\det(K_X) = 0$.

In these cases, the Gaussian vector $\boldsymbol{X}$ *does not have a probability density function. This can occur in two ways:*

- *There is an index* j, $1 \leqslant j \leqslant n$, *for which* $X_j = m_j$ *with probability 1.*
- *More generally, the "centred" variables* $X_i - m_{X_i}$, $1 \leqslant i \leqslant n$, *are not all linearly independent.*

Uncorrelation and independence

Theorem

If the random variables X_1, X_2, ..., X_n are jointly Gaussian and pairwise uncorrelated, then they are jointly independent.

Proof:

$$\mathrm{Cov}(X_i, X_j) = 0 \Longrightarrow \boldsymbol{K}_X = \mathrm{diag}(\sigma^2_{X_1}, \sigma^2_{X_2}, \ldots, \sigma^2_{X_n})$$

Uncorrelation and independence

Therefore

$$
\begin{aligned}
M_X(\omega_1, \omega_2, \ldots, \omega_n) &= \exp\left(i\boldsymbol{\omega}^t \boldsymbol{m}_X - \frac{1}{2}\boldsymbol{\omega}^t \boldsymbol{K}_X \boldsymbol{\omega}\right) \\
&= \exp\left(\sum_{k=1}^{n}\left(i\omega_k m_{X_k} - \frac{1}{2}\sigma_{X_k}^2 \omega_k^2\right)\right) \\
&= \prod_{k=1}^{n} \exp\left(i\omega_k m_{X_k} - \frac{1}{2}\sigma_{X_k}^2 \omega_k^2\right) \\
&= M_{X_1}(\omega_1) M_{X_2}(\omega_2) \cdots M_{X_n}(\omega_n)
\end{aligned}
$$

Linear combinations

Theorem

Let $\boldsymbol{X}$ be an n-dimensional Gaussian random vector, $\boldsymbol{A}$ an $m \times n$ real matrix, and $\boldsymbol{b}$ a constant real $m \times 1$ vector. Define

$$\boldsymbol{Y} = \boldsymbol{A}\boldsymbol{X} + \boldsymbol{b}$$

Then $\boldsymbol{Y}$ is an m-dimensional Gaussian random vector with

$$\boldsymbol{m}_Y = \boldsymbol{A}\,\boldsymbol{m}_X + \boldsymbol{b}, \qquad \boldsymbol{K}_Y = \boldsymbol{A}\boldsymbol{K}_X\boldsymbol{A}^t$$

- If $m \leqslant n$ and $\boldsymbol{A}$ has full rank m, the Gaussian vector $\boldsymbol{Y}$ has a density $f_Y(y_1, \ldots, y_m)$, provided that $\boldsymbol{X}$ has a density $f_X(x_1, \ldots, x_n)$.

Linear combinations

Proof: It only remains to be proved that $\boldsymbol{Y}$ is Gaussian.

$$
\begin{aligned}
M_Y(\omega_1,\omega_2,\ldots,\omega_m) &= \mathbb{E}\left(e^{i\boldsymbol{\omega}^t\boldsymbol{Y}}\right) \\
&= \mathbb{E}\left(e^{i\boldsymbol{\omega}^t(\boldsymbol{A}\boldsymbol{X}+\boldsymbol{b})}\right) = e^{i\boldsymbol{\omega}^t\boldsymbol{b}}\, M_X(\boldsymbol{\omega}^t\boldsymbol{A}) \\
&= \exp\left(i\boldsymbol{\omega}^t\boldsymbol{b}\right)\cdot\exp\left(i\,(\boldsymbol{\omega}^t\boldsymbol{A})\boldsymbol{m}_X) - \frac{1}{2}(\boldsymbol{\omega}^t\boldsymbol{A})\boldsymbol{K}_X(\boldsymbol{\omega}^t\boldsymbol{A})^t\right) \\
&= \exp\left(i\boldsymbol{\omega}^t\boldsymbol{b}\right)\cdot\exp\left(i\,\boldsymbol{\omega}^t(\boldsymbol{A}\boldsymbol{m}_X) - \frac{1}{2}\,\boldsymbol{\omega}^t(\boldsymbol{A}\boldsymbol{K}_X\boldsymbol{A}^t)\boldsymbol{\omega}\right) \\
&= \exp\left(i\,\boldsymbol{\omega}^t(\boldsymbol{A}\boldsymbol{m}_X + \boldsymbol{b}) - \frac{1}{2}\,\boldsymbol{\omega}^t(\boldsymbol{A}\boldsymbol{K}_X\boldsymbol{A}^t)\boldsymbol{\omega}\right) \\
&= \exp\left(i\,\boldsymbol{\omega}^t\boldsymbol{m}_Y - \frac{1}{2}\boldsymbol{\omega}^t\boldsymbol{K}_Y\boldsymbol{\omega}\right)
\end{aligned}
$$

Linear combinations

Theorem

The n-dimensional random vector $\boldsymbol{X} = (X_1, \ldots, X_n)^t$ *is Gaussian if and only if the 1-dimensional random variable*

$$Y = a_1 X_1 + \cdots + a_n X_n = \boldsymbol{a}^t \boldsymbol{X}$$

is Gaussian for all $\boldsymbol{a} = (a_1, \ldots, a_n)^t \in \mathbb{R}^n$.

Linear combinations

Proof: If $\boldsymbol{X}$ is Gaussian so is Y.

Reciprocally, suppose that for all $\boldsymbol{\omega}^t = (\omega_1, \ldots, \omega_n) \in \mathbb{R}^n$, the random variable $Y = \boldsymbol{\omega}^t \boldsymbol{X}$ is Gaussian.

Since

$$m_Y = \boldsymbol{\omega}^t \boldsymbol{m}_X, \qquad \sigma_Y^2 = \boldsymbol{\omega}^t \boldsymbol{K}_X \boldsymbol{\omega}$$

we have

$$\begin{aligned} M_X(\omega_1, \ldots, \omega_n) &= \mathbb{E}\left(e^{i\boldsymbol{\omega}^t \boldsymbol{X}}\right) = \mathbb{E}\left(e^{iY}\right) \\ &= M_Y(1) = \exp\left(i m_Y - \frac{1}{2}\sigma_Y^2\right) \\ &= \exp\left(i\boldsymbol{\omega}^t \boldsymbol{m}_X - \frac{1}{2}\boldsymbol{\omega}^t \boldsymbol{K}_X \boldsymbol{\omega}\right) \end{aligned}$$

Linear combinations

Let $\boldsymbol{K}$ be a positive-semidefinite symmetric matrix and set:

$$\boldsymbol{C}\boldsymbol{K}\boldsymbol{C}^t = \boldsymbol{D} = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n), \quad \lambda_i \geqslant 0$$

$$\boldsymbol{D}^{1/2} = \mathrm{diag}(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \ldots, \sqrt{\lambda_n})$$

$$\boldsymbol{K}^{1/2} = \boldsymbol{C}^t \boldsymbol{D}^{1/2} \boldsymbol{C}$$

Then $\boldsymbol{D}^{1/2}\boldsymbol{D}^{1/2} = \boldsymbol{D}$, the matrix $\boldsymbol{K}^{1/2}$ is symmetric, and

$$\begin{aligned}
\boldsymbol{K}^{1/2}\boldsymbol{K}^{1/2} &= (\boldsymbol{C}^t \boldsymbol{D}^{1/2} \boldsymbol{C})(\boldsymbol{C}^t \boldsymbol{D}^{1/2} \boldsymbol{C}) \\
&= \boldsymbol{C}^t \boldsymbol{D}^{1/2} (\boldsymbol{C}\boldsymbol{C}^t) \boldsymbol{D}^{1/2} \boldsymbol{C} \\
&= \boldsymbol{C}^t (\boldsymbol{D}^{1/2}\boldsymbol{D}^{1/2}) \boldsymbol{C} = \boldsymbol{C}^t \boldsymbol{D} \boldsymbol{C} = \boldsymbol{K}
\end{aligned}$$

Linear combinations

Proposition

Let $\boldsymbol{Z}$ be a random vector with independent components $Z_1, Z_2, \ldots, Z_n$, each distributed as $\mathrm{N}(0,1)$. Let $\boldsymbol{K}$ be an $n \times n$ positive-semidefinite symmetric matrix, and $\boldsymbol{m}$ an $n \times 1$ real constant vector. If

$$\boldsymbol{X} = \boldsymbol{K}^{1/2}\boldsymbol{Z} + \boldsymbol{m},$$

then $\boldsymbol{X} \sim \mathrm{N}(\boldsymbol{m}, \boldsymbol{K})$.

Proof: Notice that $\mathbb{E}(\boldsymbol{X}) = \boldsymbol{K}^{1/2}\,\mathbb{E}(\boldsymbol{Z}) + \boldsymbol{m} = \boldsymbol{m}$. It only remains to be proved that the covariance matrix of $\boldsymbol{X}$ is $\boldsymbol{K}$. Indeed,

$$\boldsymbol{K}_X = \boldsymbol{K}^{1/2}\,\boldsymbol{K}_Z\,(\boldsymbol{K}^{1/2})^t = \boldsymbol{K}^{1/2}\,\boldsymbol{I}\,(\boldsymbol{K}^{1/2})^t = \boldsymbol{K}^{1/2}\boldsymbol{K}^{1/2} = \boldsymbol{K}$$

The multivariate Gaussian density

Let $\boldsymbol{Z} \sim \mathrm{N}(\mathbf{0}, \boldsymbol{I})$ and consider the transformation

$$\boldsymbol{X} = \boldsymbol{K}^{1/2}\boldsymbol{Z} + \boldsymbol{m},$$

where $\boldsymbol{K}$ is an $n \times n$ positive-definite symmetric matrix (hence $\det(\boldsymbol{K}) > 0$).

We know that $\boldsymbol{X} \sim \mathrm{N}(\boldsymbol{m}, \boldsymbol{K})$. Let us find $f_X(x_1, x_2, \ldots, x_n)$.

We have

$$\begin{aligned} f_Z(z_1, z_2, \ldots, z_n) &= f_{Z_1}(z_1) f_{Z_2}(z_2) \cdots f_{Z_n}(z_n) \\ &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}}\, e^{-\frac{1}{2}\, z_i^2} = \frac{1}{(2\pi)^{n/2}}\, e^{-\frac{1}{2}\boldsymbol{z}^t\boldsymbol{z}} \end{aligned}$$

The multivariate Gaussian density

Let $\boldsymbol{K}^{-1/2} = (\boldsymbol{K}^{1/2})^{-1}$. The matrix $\boldsymbol{K}^{-1/2}$ is symmetric and, moreover,

$$\boldsymbol{K}^{-1/2}\boldsymbol{K}^{-1/2} = \boldsymbol{K}^{-1}$$

- The equation $\boldsymbol{x} = \boldsymbol{K}^{1/2}\boldsymbol{z} + \boldsymbol{m}$ has a unique solution:

$$\boldsymbol{z} = \boldsymbol{K}^{-1/2}(\boldsymbol{x} - \boldsymbol{m})$$

- The Jacobian determinant of the transformation is

$$J(z_1, z_2, \ldots, z_n) = \det(\boldsymbol{K}^{1/2}) = \sqrt{\det(\boldsymbol{K})}$$

The multivariate Gaussian density

In this way, by the transformation theorem we have

$$
\begin{aligned}
f_X(x_1, x_2, \ldots, x_n) &= \left.\frac{f_Z(z_1, z_2, \ldots, z_n)}{|J(z_1, z_2, \ldots, z_n)|}\right|_{\boldsymbol{z}=\boldsymbol{K}^{-1/2}(\boldsymbol{x}-\boldsymbol{m})} \\
&= \frac{1}{(2\pi)^{n/2}\sqrt{\det(\boldsymbol{K})}} e^{-\frac{1}{2}\left(\boldsymbol{K}^{-1/2}(\boldsymbol{x}-\boldsymbol{m})\right)^t\left(\boldsymbol{K}^{-1/2}(\boldsymbol{x}-\boldsymbol{m})\right)} \\
&= \frac{1}{(2\pi)^{n/2}\sqrt{\det(\boldsymbol{K})}} e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{m})^t\left(\boldsymbol{K}^{-1/2}\right)^t\boldsymbol{K}^{-1/2}(\boldsymbol{x}-\boldsymbol{m})} \\
&= \frac{1}{(2\pi)^{n/2}\sqrt{\det(\boldsymbol{K})}} e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{m})^t\boldsymbol{K}^{-1}(\boldsymbol{x}-\boldsymbol{m})}
\end{aligned}
$$

The multivariate Gaussian density

Proposition

Let $\boldsymbol{X} \sim \mathrm{N}(\boldsymbol{m}, \boldsymbol{K})$*, where* $\det(\boldsymbol{K}) > 0$*. Then* $\boldsymbol{X}$ *has a density function* f_X *such that*

$$f_X(x_1, x_2, \ldots, x_n) = \frac{1}{(2\pi)^{n/2}\sqrt{\det(\boldsymbol{K})}} \cdot \exp\left(-\frac{1}{2}(\boldsymbol{x} - \boldsymbol{m})^t \, \boldsymbol{K}^{-1}(\boldsymbol{x} - \boldsymbol{m})\right)$$

The multivariate Gaussian density

For instance, for $n = 2$ we obtain:

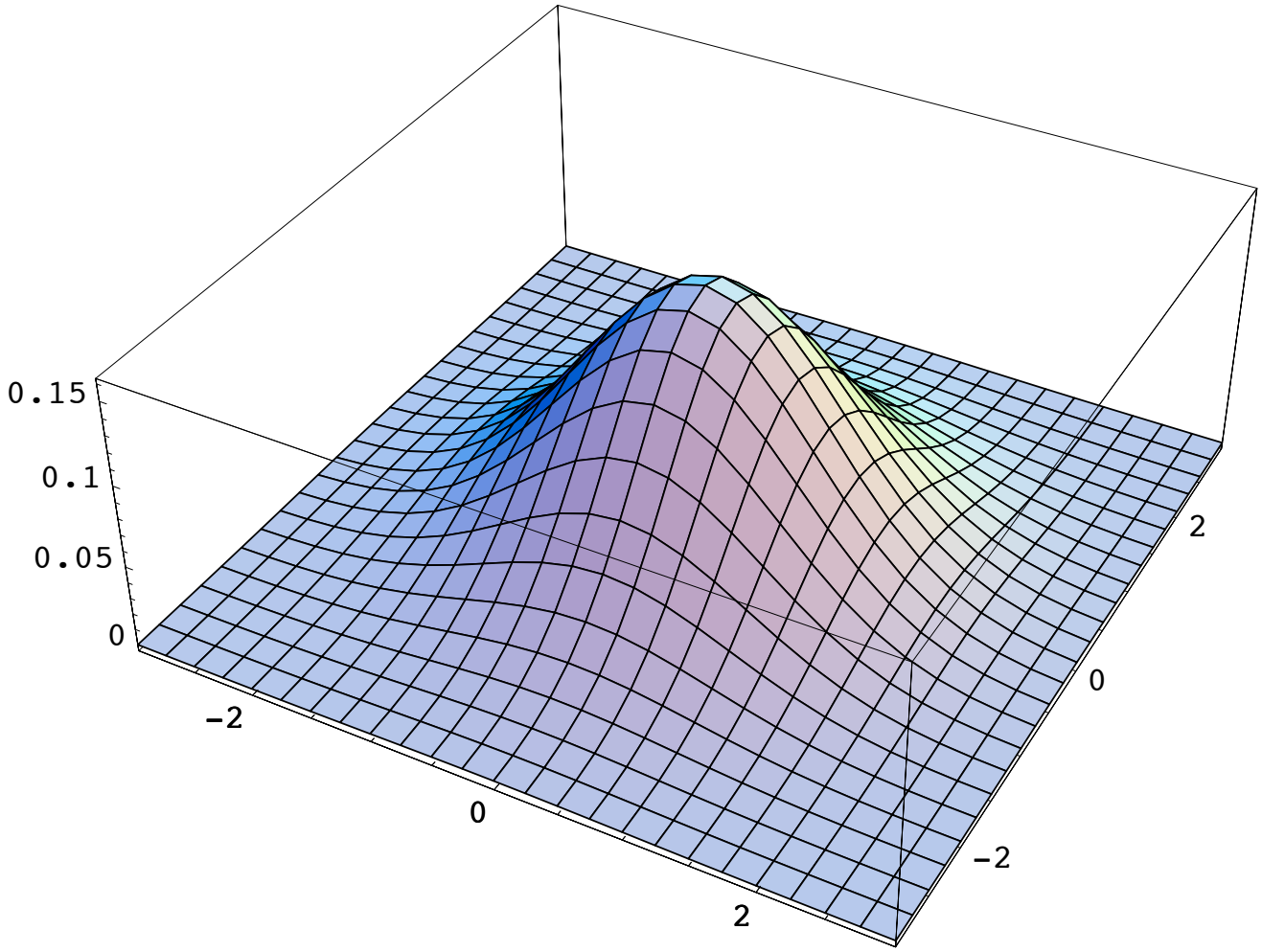
$$f_{XY}(x, y) = \frac{1}{2\pi\,\sqrt{1-\rho^2}\,\sigma_X\sigma_Y}\,\exp\left(-\frac{1}{2}\cdot\frac{1}{1-\rho^2}\cdot a(x,y)\right),$$

where

$$a(x, y) = \left(\frac{x - m_X}{\sigma_X}\right)^2 - 2\rho\,\frac{x - m_X}{\sigma_X}\cdot\frac{y - m_Y}{\sigma_Y} + \left(\frac{y - m_Y}{\sigma_Y}\right)^2$$

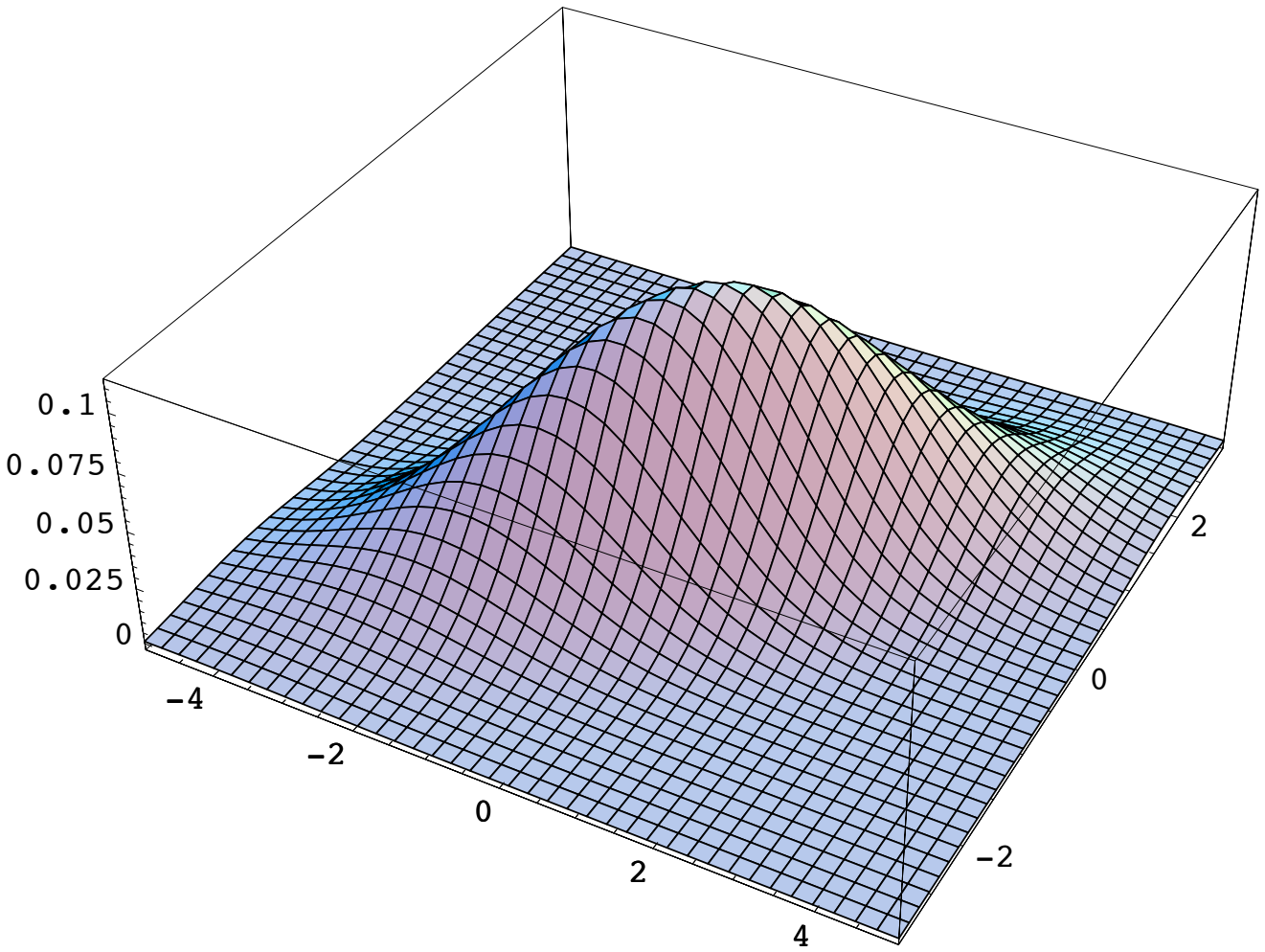
The multivariate Gaussian density

$$\sigma_X = \sigma_Y \qquad \rho = 0$$



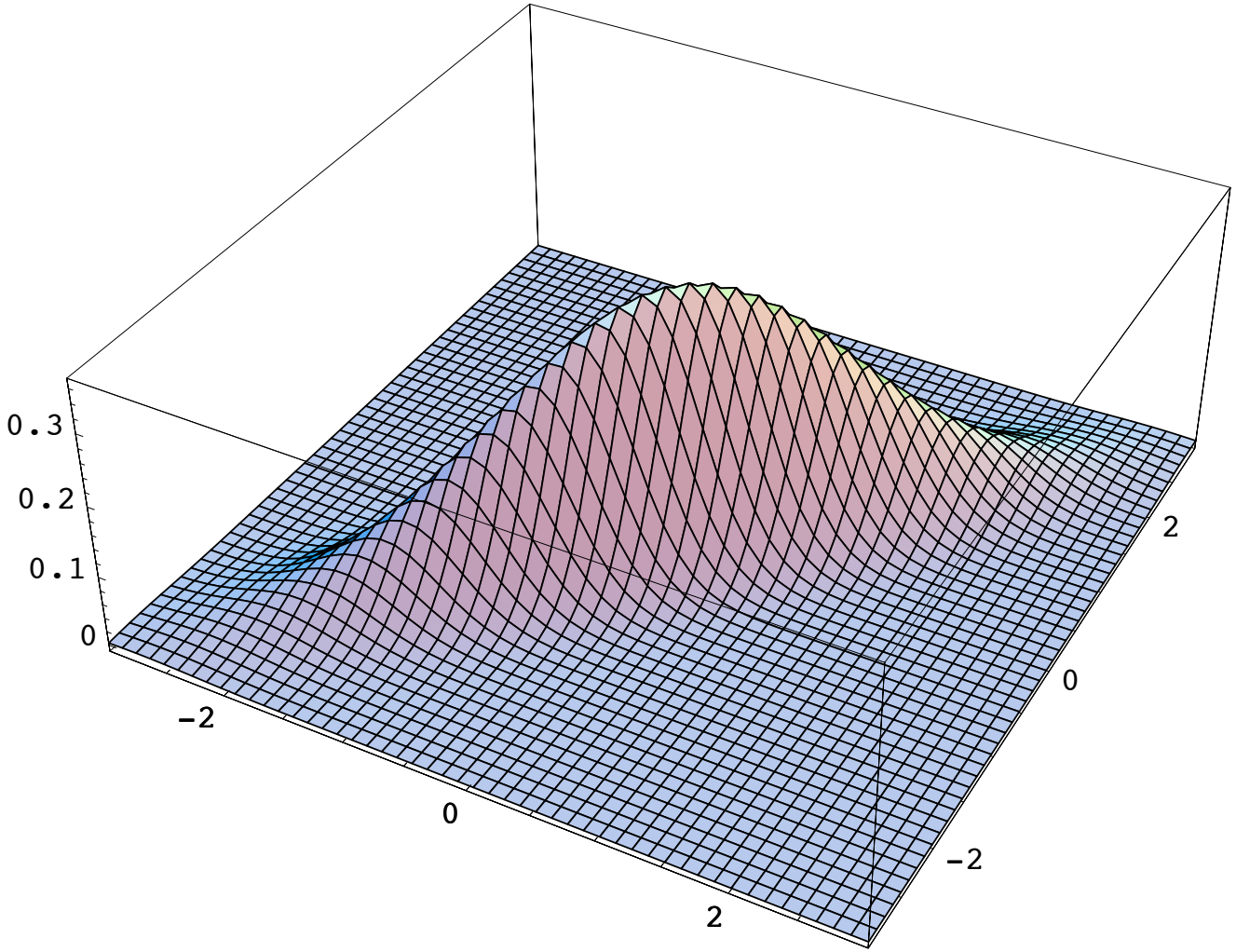
The multivariate Gaussian density

$$\sigma_X = 2\sigma_Y \qquad \rho = 0.7$$



The multivariate Gaussian density

$$\sigma_X = \sigma_Y \qquad \rho = 0.9$$



Conditional densities

Let X, Y be jointly Gaussian. Then,

$$f_{Y|X}(y|X=x) = \frac{f_{XY}(x,y)}{f_X(x)}$$
$$= \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}\,\sigma_Y} \exp\left(-\frac{1}{2}\left(\frac{y-m_{Y|X}}{\sigma_{Y|X}}\right)^2\right)$$

- $m_{Y|X}$ is the expected value of Y given X:

$$m_{Y|X} = \mathbb{E}(Y|X=x) = \rho\,\frac{\sigma_Y}{\sigma_X}(x-m_X) + m_Y$$

- $\sigma^2_{Y|X} = (1-\rho^2)\,\sigma^2_Y$.

Additional results

Theorem

Let $\boldsymbol{X} \sim \mathrm{N}(\boldsymbol{m}, \boldsymbol{K})$ *with* $\det(\boldsymbol{K}) > 0$. *Then the random variable*

$$(\boldsymbol{X} - \boldsymbol{m})^t \boldsymbol{K}^{-1} (\boldsymbol{X} - \boldsymbol{m})$$

follows a $\chi^2(n)$*-distribution, where* n *is de dimension of* $\boldsymbol{X}$.

For instance, for $n = 2$, the random variable

$$\frac{1}{1-\rho^2}\left(\left(\frac{X_1 - m_1}{\sigma_1}\right)^2 - 2\rho\,\frac{X_1 - m_1}{\sigma_1}\cdot\frac{X_2 - m_2}{\sigma_2} + \left(\frac{X_2 - m_2}{\sigma_2}\right)^2\right)$$

is $\chi^2(2)$-distributed.

Additional results

Proof: Consider $\boldsymbol{Y} = \boldsymbol{K}^{-1/2}(\boldsymbol{X} - \boldsymbol{m})$. Then,

$$\mathbb{E}(\boldsymbol{Y}) = \mathbb{E}\left(\boldsymbol{K}^{-1/2}(\boldsymbol{X} - \boldsymbol{m})\right) = \boldsymbol{K}^{-1/2}\mathbb{E}((\boldsymbol{X} - \boldsymbol{m})) = \boldsymbol{0}$$

$$\begin{aligned}
\boldsymbol{K}_Y &= \mathbb{E}\left(\boldsymbol{Y}\boldsymbol{Y}^t\right) = \mathbb{E}\left(\boldsymbol{K}^{-1/2}(\boldsymbol{X} - \boldsymbol{m})(\boldsymbol{X} - \boldsymbol{m})^t(\boldsymbol{K}^{-1/2})^t\right) \\
&= \mathbb{E}\left(\boldsymbol{K}^{-1/2}(\boldsymbol{X} - \boldsymbol{m})(\boldsymbol{X} - \boldsymbol{m})^t\boldsymbol{K}^{-1/2}\right) \\
&= \boldsymbol{K}^{-1/2}\,\mathbb{E}\left((\boldsymbol{X} - \boldsymbol{m})(\boldsymbol{X} - \boldsymbol{m})^t\right)\boldsymbol{K}^{-1/2} \\
&= \boldsymbol{K}^{-1/2}\boldsymbol{K}\boldsymbol{K}^{-1/2} = (\boldsymbol{K}^{-1/2}\boldsymbol{K}^{1/2})(\boldsymbol{K}^{1/2}\boldsymbol{K}^{-1/2}) = \boldsymbol{I}
\end{aligned}$$

Additional results

That is, $\boldsymbol{Y} = \boldsymbol{K}^{-1/2}(\boldsymbol{X} - \boldsymbol{m}) \sim \mathrm{N}(\boldsymbol{0}, \boldsymbol{I})$. Therefore,

$$\begin{aligned}
&(\boldsymbol{X} - \boldsymbol{m})^t \boldsymbol{K}^{-1}(\boldsymbol{X} - \boldsymbol{m}) \\
&\quad = (\boldsymbol{X} - \boldsymbol{m})^t \boldsymbol{K}^{-1/2} \boldsymbol{K}^{-1/2}(\boldsymbol{X} - \boldsymbol{m}) \\
&\quad = \boldsymbol{Y}^t \boldsymbol{Y} = \sum_{i=1}^{n} Y_i^2
\end{aligned}$$

is a (1-dimensional) random variable following a $\chi^2(n)$-distribution, because the variables Y_i, $1 \leqslant i \leqslant n$, are independent and $\mathrm{N}(0, 1)$-distributed.

Additional results

Theorem

Let $\boldsymbol{X} \sim \mathrm{N}(\boldsymbol{m}, \boldsymbol{K})$ *and set* $\boldsymbol{Y} = \boldsymbol{C}\boldsymbol{X}$*, where* $\boldsymbol{C}$ *is an orthogonal matrix such that* $\boldsymbol{C}\boldsymbol{K}\boldsymbol{C}^t = \boldsymbol{D} = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$.

Then, $\boldsymbol{Y} \sim \mathrm{N}(\boldsymbol{C}\boldsymbol{m}, \boldsymbol{D})$.

In particular, the components of $\boldsymbol{Y}$ *are independent and*

$$\mathrm{Var}(Y_k) = \lambda_k, \quad k = 1, \ldots, n$$

Remark: It may occur that some eigenvalue is equal to 0, in which case the corresponding component is degenerate.

Additional results

Theorem

Let $\boldsymbol{X} \sim \mathrm{N}(\boldsymbol{m}, \sigma^2 \boldsymbol{I})$*, where* $\sigma^2 > 0$*. Let* $\boldsymbol{C}$ *be an arbitrary orthogonal matrix, and set* $\boldsymbol{Y} = \boldsymbol{C}\boldsymbol{X}$*. Then,* $\boldsymbol{Y} \sim \mathrm{N}(\boldsymbol{C}\boldsymbol{m}, \sigma^2 \boldsymbol{I})$*.*

In particular, the components of $\boldsymbol{Y}$ *are independent Gaussian random variables with the same variance* σ^2*.*

Proof:

$$\boldsymbol{K}_Y = \boldsymbol{C}\boldsymbol{K}_X\boldsymbol{C}^t = \boldsymbol{C}(\sigma^2\boldsymbol{I})\boldsymbol{C}^t = \sigma^2\boldsymbol{C}\boldsymbol{C}^t = \sigma^2\boldsymbol{I}$$

Additional results

Theorem

Let $\boldsymbol{X} \sim \mathrm{N}(\boldsymbol{m}, \boldsymbol{K})$*, and suppose that* $\boldsymbol{K}$ *can be partitioned (possibly after reordering the components) as follows:*

$$\boldsymbol{K} = \begin{pmatrix} \boldsymbol{K}_1 & & & \\ & \boldsymbol{K}_2 & & 0 \\ & & \cdots & \\ & 0 & & \cdots \\ & & & & \boldsymbol{K}_p \end{pmatrix}.$$

Then, $\boldsymbol{X}$ *can be partitioned into vectors* $\boldsymbol{X}^{(1)}$*,* $\boldsymbol{X}^{(2)}$*, ...,* $\boldsymbol{X}^{(p)}$*, where* $\boldsymbol{K}_i$ *is the covariance matrix of* $\boldsymbol{X}^{(i)}$*,* $i = 1, 2, \ldots, p$*, and in such a way that the random vectors* $\boldsymbol{X}^{(1)}$*,* $\boldsymbol{X}^{(2)}$*, ...,* $\boldsymbol{X}^{(p)}$ *are independent.*

Example

Let $\boldsymbol{X} = (X_1, X_2, X_3)^t$ be a Gaussian vector with $\boldsymbol{m} = (0, 0, 0)^t$ and

$$\boldsymbol{K} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 4 \\ 0 & 4 & 9 \end{pmatrix}$$

Then X_1 and (X_2, X_3) are independent.

Example

Indeed,

$$
\begin{aligned}
M_X(\omega_1, \omega_2, \omega_3) &= \exp\left(i\boldsymbol{\omega}^t \boldsymbol{m} - \frac{1}{2}\boldsymbol{\omega}^t \boldsymbol{K}\boldsymbol{\omega}\right) \\
&= \exp\left(-\frac{1}{2}(\omega_1, \omega_2, \omega_3)\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 4 \\ 0 & 4 & 9 \end{pmatrix}\begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}\right) \\
&= \exp\left(-\frac{1}{2}\omega_1^2\right) \cdot \exp\left(-\frac{1}{2}(\omega_2, \omega_3)\begin{pmatrix} 2 & 4 \\ 4 & 9 \end{pmatrix}\begin{pmatrix} \omega_2 \\ \omega_3 \end{pmatrix}\right) \\
&= M_{X_1}(\omega_1) \cdot M_{X_2 X_3}(\omega_2, \omega_3)
\end{aligned}
$$